SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO4 ASSESSMENT TOOL 1 – Test Case Design Assignment

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO4	Assessment:	Assessment Tool 1 – Test Case Design Assignment
Weightage:	20%	Total Marks:	25
CO4 Statement:	Implement robust testing, quality assurance, and DevOps practices, emphasizing security, reliability, and ethics		
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Date:	21/08/2026	Duration:	60 minutes

Code of Conduct

*I **DHANUSHREE S/192421218** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: Dhanushree S

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Requirement & Test Scenario Identification(15%)	Clearly identifies all relevant functional/non-functional requirements and comprehensive test scenarios	Identifies most requirements and scenarios	Identifies basic requirements and scenarios	Requirements/scenarios are incomplete or unclear
Test Case Design & Completeness(25%)	Comprehensive test cases covering normal, boundary, invalid, and exceptional conditions	Covers most important conditions with minor gaps	Covers basic conditions but misses several important cases	Test cases are very limited or incomplete
Test Data & Expected Results(20%)	Appropriate test data with precise, measurable expected results for every test case	Mostly appropriate test data and expected results	Some test data/results are unclear or incomplete	Test data and expected results are largely missing/incorrect
Application of Test Design Techniques(15%)	Correctly applies multiple techniques such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, and State Transition	Correctly applies at least two relevant techniques	Applies one technique with limited justification	Techniques are not applied or incorrectly applied
Traceability & Test Coverage(15%)	Excellent mapping between requirements, scenarios, and test cases with strong coverage	Good traceability and coverage with minor gaps	Partial traceability and moderate coverage	Poor/no traceability and inadequate coverage
Documentation & Presentation(10%)	Well-structured, clear, professional, consistent, and easy to understand	Well-organized with minor presentation issues	Adequately organized but has some clarity/formatting issues	Poorly organized, unclear, or incomplete documentation

AI-BASED CROP DISEASE DETECTION AND FARMER ADVISORY PLATFORM

1. Introduction

The AI-Based Crop Disease Detection and Farmer Advisory Platform is a mobile and web-based system that enables farmers to capture or upload images of crop leaves, receive an AI-generated diagnosis of possible plant diseases, and obtain actionable advisory recommendations such as treatment options, pesticide dosage, and preventive measures. The platform also integrates weather-based alerts, multilingual support for regional accessibility, diagnosis history tracking, and an administrative dashboard for managing the disease knowledge base and monitoring regional disease trends. Given the system's direct impact on farmers' livelihoods and crop yield, rigorous testing of its functional accuracy, reliability, security, and usability is essential.

2. Purpose and Scope of Testing

This document defines the requirement analysis, test scenarios, test design techniques, and detailed test cases required to verify the correctness, functionality, reliability, and usability of the AI-Based Crop Disease Detection and Farmer Advisory Platform, in accordance with the instructions of the CO4 Assessment Tool 1. The scope covers the farmer-facing mobile application, the AI disease-detection and advisory engine, backend services (notifications, weather integration, database), and the admin dashboard.

3. Test Environment and Tools

The following environment and tools are assumed for designing and, subsequently, executing the test cases documented in this assignment.

- Mobile Platform: Android 8.0 and above (primary), with a companion responsive web dashboard for administrators.

- Backend: REST API-based microservices hosted on a cloud environment (e.g., AWS/GCP) with a relational/NoSQL database for farmer, disease, and diagnosis records.
- AI Model: Convolutional Neural Network (CNN)-based image classification model trained on a labeled crop-disease image dataset.
- Test Management: A spreadsheet or test-management tool (e.g., TestRail, Excel) to log test case execution, actual results, and defects.
- Defect Tracking: An issue-tracking tool (e.g., Jira) to log and track defects identified during execution.
- Network Simulation: Tools/emulators to simulate 2G/3G/offline conditions for NFR7 and NFR10 verification.
- Browsers/Devices: Chrome and a mid/low-range Android device for cross-compatibility checks.

4. Assumptions and Constraints

The following assumptions and constraints were considered while identifying requirements and designing the test cases in this document.

- The AI disease-detection model is assumed to be already trained and deployed as a callable service; model training itself is outside the scope of this test design.
- Farmers are assumed to have access to a basic smartphone with a working camera and intermittent internet connectivity.
- Weather data is sourced from a third-party API whose availability is assumed but not directly controlled by the platform team.
- Only five regional languages are assumed to be supported in the current release; additional languages are out of scope.
- Test execution (filling Actual Result and Status columns) is planned as a subsequent phase after this design document is reviewed and approved.

5. System Architecture Overview

The diagram below illustrates the high-level architecture of the platform, showing how the farmer's mobile application interacts with the backend gateway, the AI disease-detection engine, the advisory recommendation engine, supporting services (weather, notifications), and the admin dashboard.

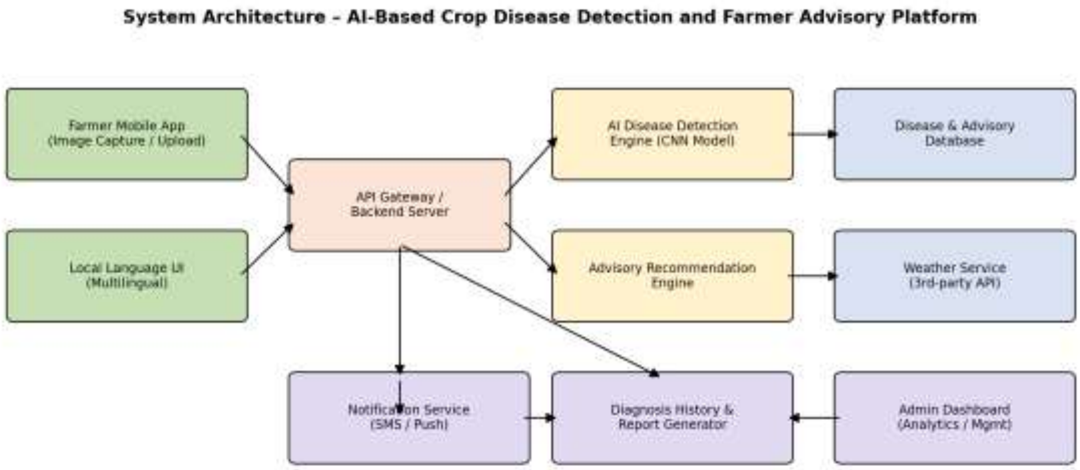

Figure 1: High-level system architecture of the platform

6. Requirement Identification

6.1 Functional Requirements

The following functional requirements were identified from the assigned problem statement and form the basis for test scenario and test case design.

ID	Functional Requirement	Priority
FR1	The system shall allow farmers to register and log in using mobile number / email and OTP or password.	High
FR2	The system shall allow farmers to capture or upload crop leaf images for disease detection.	High
FR3	The system shall analyze the uploaded image using an AI model and return the detected disease with a confidence score.	High
FR4	The system shall provide advisory recommendations (treatment, pesticide, dosage) based on the detected disease.	High
FR5	The system shall allow farmers to search and view disease information from the knowledge database.	Medium
FR6	The system shall send weather-based crop advisory alerts to registered farmers.	Medium

ID	Functional Requirement	Priority
FR7	The system shall support multiple regional languages in the user interface.	Medium
FR8	The system shall maintain diagnosis history and allow farmers to generate/download reports.	Medium
FR9	The system shall allow farmers to submit feedback/rating on the advisory received.	Low
FR10	The system shall provide an admin dashboard to manage the disease database, users, and view analytics.	High
FR11	The system shall queue and retry image uploads automatically under low-bandwidth or offline conditions.	Medium
FR12	The system shall deliver notifications to farmers via SMS and push notification.	Medium

6.2 Non-Functional Requirements

Non-functional requirements define the quality attributes the platform must satisfy, including performance, security, usability, and reliability, which are equally critical to test given the platform's use by farmers in varied network and literacy conditions.

ID	Category	Non-Functional Requirement
NFR1	Performance	The system shall return a disease detection result within 5 seconds under normal network conditions.
NFR2	Usability	The interface shall be simple and icon-driven to support farmers with low digital/text literacy.
NFR3	Reliability	The system shall maintain at least 99% uptime during the crop advisory season.
NFR4	Security	All farmer data, images, and credentials shall be encrypted in transit and at rest; access shall require authentication.
NFR5	Scalability	The system shall support at least 10,000 concurrent image-upload requests without performance degradation.
NFR6	Accuracy	The AI model shall achieve a minimum detection precision and recall of 85% on the validation dataset.
NFR7	Compatibility	The mobile application shall function correctly on low-end Android devices (Android 8.0 and above, 2 GB RAM).

ID	Category	Non-Functional Requirement
NFR8	Localization	The platform shall correctly render and function in at least five regional languages.
NFR9	Maintainability	The disease/advisory database shall be updatable by administrators without requiring application redeployment.
NFR10	Availability (Low Network)	The system shall queue requests locally and sync automatically once connectivity is restored.

7. Test Scenario Identification

Based on the functional and non-functional requirements above, the following high-level test scenarios were identified. Each scenario is later broken down into detailed test cases covering normal, boundary, invalid, and exceptional conditions.

Scenario ID	Linked Requirement(s)	Test Scenario Description
TS01	FR1	Verify farmer registration and login functionality
TS02	FR2	Verify crop image capture and upload functionality
TS03	FR3	Verify AI-based disease detection and confidence scoring
TS04	FR4	Verify advisory recommendation generation based on detected disease
TS05	FR5	Verify disease database search and view functionality
TS06	FR6	Verify weather-based alert notifications
TS07	FR7	Verify multilingual interface switching
TS08	FR8	Verify diagnosis history logging and report generation
TS09	FR9	Verify feedback and rating submission
TS10	FR10	Verify admin dashboard operations (user/database management, analytics)
TS11	FR11 / NFR10	Verify system behavior under offline / low-bandwidth conditions
TS12	FR12	Verify SMS and push notification delivery
TS13	NFR4	Verify data security and access control

Scenario ID	Linked Requirement(s)	Test Scenario Description
TS14	NFR1 / NFR5	Verify system performance and scalability under load
TS15	NFR6	Verify AI model detection accuracy against known samples

8. Application of Test Design Techniques

To ensure comprehensive and systematic coverage, four standard test design techniques were applied while designing the test cases in Section 7.

8.1 Equivalence Partitioning (EP)

Input fields were divided into valid and invalid partitions so that one representative test case from each partition provides adequate coverage without redundant testing.

Input Field	Valid Partition	Invalid Partition
Mobile Number	Exactly 10 numeric digits	<10 digits, >10 digits, or contains letters
Image File Format	JPG, JPEG, PNG	PDF, BMP, GIF, TXT
Feedback Rating	Integers 1 to 5	0, negative numbers, or values > 5

8.2 Boundary Value Analysis (BVA)

Since defects commonly occur at the edges of input ranges, boundary values (just inside, at, and just outside each limit) were specifically tested for size-, length-, and threshold-based fields.

Field	Min Boundary	Max Boundary	Test Points
Password Length	8 characters	16 characters	7, 8, 16, 17
Image Upload Size	1 KB	5 MB	0 KB, 5 MB, 5.1 MB
AI Confidence Threshold	60%	100%	59%, 60%, 100%
Feedback Rating	1	5	0, 1, 5, 6

8.3 Decision Table Testing

The advisory recommendation logic depends on a combination of disease type and severity. A decision table was used to derive test cases that verify every relevant rule combination produces the correct advisory output.

Disease Type	Severity	Condition	Advisory Action
Fungal	High	Rule 1	Fungicide + urgent isolation
Fungal	Low	Rule 2	Preventive spray + monitoring
Bacterial	High	Rule 3	Bactericide + crop rotation advice
Unidentified	N/A	Rule 4	Refer to local agriculture officer

8.4 State Transition Testing

The lifecycle of a diagnosis request moves through multiple states from submission to completion or failure. State transition testing was applied to verify that each valid transition behaves correctly and that invalid transitions are handled gracefully.

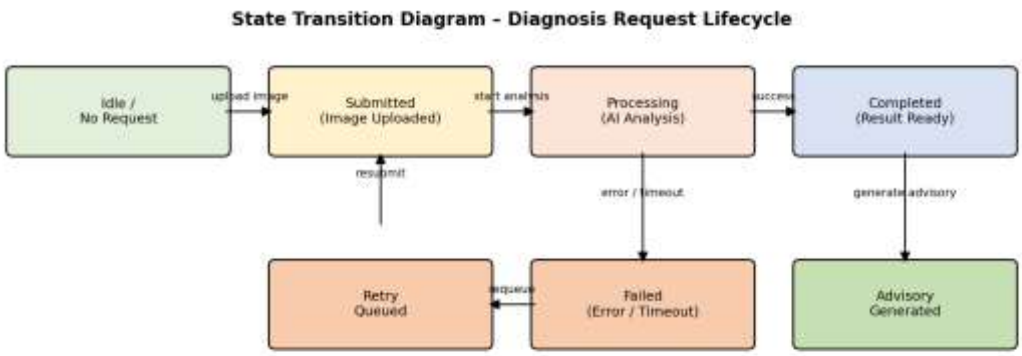

Figure 2: State transition diagram for the diagnosis request lifecycle

9. Detailed Test Cases

The following table presents the detailed test cases derived from the test scenarios in Section 5 and the design techniques in Section 6. Each test case covers normal, boundary, invalid, or exceptional conditions as applicable. The 'Actual Result' and 'Status' columns are to be completed during the execution phase; they are marked as Not Executed / Pending in this design document since execution occurs after this design is reviewed and approved.

TC ID	Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC01	TS01	Register with valid mobile number, name, and password	Name: Ravi Kumar; Mobile: 9876543210; Password: Farm@123	Account created successfully; OTP sent for verification	Not Executed	Pending
TC02	TS01	Register with an already registered mobile number	Mobile: 9876543210 (existing)	System displays 'Mobile number already registered' error	Not Executed	Pending
TC03	TS01	Register with invalid mobile number format (Equivalence Partitioning - invalid class)	Mobile: 98765 (5 digits)	System rejects with 'Enter a valid 10-digit mobile number'	Not Executed	Pending
TC04	TS01	Register with password below minimum boundary	Password: Fa1@ (4 characters, min=8)	System rejects; 'Password must be	Not Executed	Pending

		(Boundary Value Analysis)		at least 8 characters'		
TC05	TS01	Login with valid registered credentials	Mobile: 9876543210; Password: Farm@123	Farmer is redirected to home dashboard	Not Executed	Pending
TC06	TS01	Login with incorrect password	Mobile: 9876543210; Password: Wrong@123	System displays 'Incorrect password' and denies access	Not Executed	Pending
TC07	TS01	Login with unregistered mobile number	Mobile: 9123456780	System displays 'Account not found'	Not Executed	Pending
TC08	TS01	Verify account lockout after repeated failed logins (Boundary Value Analysis)	5 consecutive wrong password attempts	Account is temporarily locked after the 5th attempt	Not Executed	Pending
TC09	TS02	Upload a valid crop leaf image in supported format	Image: leaf01.jpg, 2 MB	Image uploads successfully and is queued for analysis	Not Executed	Pending

TC10	TS02	Upload image at maximum allowed size boundary (BVA)	Image size: exactly 5 MB (max limit)	Image is accepted for upload	Not Executed	Pending
TC11	TS02	Upload image exceeding maximum size boundary (BVA)	Image size: 5.1 MB	System rejects with 'File size exceeds 5 MB limit'	Not Executed	Pending
TC12	TS02	Upload file in an unsupported format (Equivalence Partitioning - invalid class)	File: leaf_scan.pdf	System rejects with 'Unsupported file format. Use JPG/PNG'	Not Executed	Pending
TC13	TS02	Upload a blurred / low-quality image	Image: blurred_leaf.jpg	System warns 'Image quality too low, please retake photo'	Not Executed	Pending
TC14	TS02	Capture image directly via in-app camera	Live camera capture of a tomato leaf	Captured image is submitted for analysis	Not Executed	Pending
TC15	TS02	Attempt upload with no image selected	No file attached	System displays 'Please select or	Not Executed	Pending

				capture an image'		
TC16	TS02	Upload an image with no crop/leaf content (invalid input class)	Image: random_object.jpg	System returns 'No crop leaf detected in image'	Not Executed	Pending
TC17	TS03	Detect disease from a clear image of an infected leaf	Image: tomato_blight.jpg	System returns disease name 'Late Blight' with confidence ≥ 85%	Not Executed	Pending
TC18	TS03	Detect a healthy leaf with no disease	Image: healthy_leaf.jpg	System returns 'No disease detected - Healthy crop'	Not Executed	Pending
TC19	TS03	Verify confidence score at minimum acceptable boundary (BVA)	Model confidence output: exactly 60% (threshold)	Result is displayed with a 'Low Confidence' advisory note	Not Executed	Pending
TC20	TS03	Verify confidence score just below	Model confidence output: 59%	System prompts 'Please upload a	Not Executed	Pending

		acceptance boundary (BVA)		clearer image for accurate results'		
TC21	TS03	Detect disease under poor lighting conditions	Image: leaf_lowlight.jpg	System still returns a result or requests image retake if unreliabl e	Not Executed	Pending
TC22	TS03	Verify response time for detection result (Performance)	Standard 2 MB image on 4G network	Result is returned within 5 seconds	Not Executed	Pending
TC23	TS03	Detect disease for a crop type outside the trained model scope (invalid class)	Image: unsupported_crop.jp g	System returns 'Crop type not supported currently'	Not Executed	Pending
TC24	TS04	Generate advisory for High severity + Fungal disease (Decision Table Rule 1)	Disease: Late Blight; Severity: High; Type: Fungal	Advisory recomme nds fungicide , dosage, and urgent isolation of plants	Not Executed	Pending
TC25	TS04	Generate advisory for Low severity + Fungal disease	Disease: Early Blight; Severity: Low; Type: Fungal	Advisory recomme nds preventiv	Not Executed	Pending

		(Decision Table Rule 2)		e fungicide spray and monitorin g		
TC26	TS04	Generate advisory for High severity + Bacterial disease (Decision Table Rule 3)	Disease: Bacterial Wilt; Severity: High; Type: Bacterial	Advisory recomme nds bactericid e and crop rotation guidance	Not Executed	Pending
TC27	TS04	Generate advisory when disease is Unidentified (Decision Table Rule 4)	Disease: Unidentified; Severity: N/A	Advisory recomme nds consultin g a local agricultur e officer	Not Executed	Pending
TC28	TS04	Verify advisory includes dosage and safety instructions	Disease: Powdery Mildew; Severity: Medium	Advisory shows pesticide name, dosage, and safety precautio ns	Not Executed	Pending
TC29	TS05	Search disease database using a valid disease name	Search term: 'Blight'	Matching disease records are displayed with details	Not Executed	Pending

TC30	TS05	Search using a term with no matching records	Search term: 'xyzabc'	System displays 'No matching records found'	Not Executed	Pending
TC31	TS05	Search with an empty query string	Search term: '' (blank)	System prompts 'Please enter a search term'	Not Executed	Pending
TC32	TS06	Receive weather alert for a registered farmer's location	Location: Coimbatore; Forecast: Heavy rainfall	Farmer receives a relevant crop-protection weather alert	Not Executed	Pending
TC33	TS06	Verify no alert is sent when weather conditions are normal	Forecast: Clear, no adverse conditions	No alert notification is generated	Not Executed	Pending
TC34	TS06	Verify alert delivery when location services are disabled	Location permission: Denied	System prompts farmer to enable location for weather alerts	Not Executed	Pending
TC35	TS07	Switch application language from English to Tamil	Language setting: Tamil	All UI labels and advisory	Not Executed	Pending

				text are displayed in Tamil		
TC36	TS07	Verify unsupported language selection is not available	Language list check	Only the supported five regional language s are listed	Not Executed	Pending
TC37	TS08	View diagnosis history after multiple scans	3 previous diagnosis records exist	History list displays all past diagnoses with date and result	Not Executed	Pending
TC38	TS08	Generate and download a PDF report of a diagnosis	Select diagnosis record #1023	PDF report is generated with disease, advisory, and date	Not Executed	Pending
TC39	TS08	View diagnosis history when no records exist (boundary/empt y state)	New account with zero scans	System displays 'No diagnosis history available'	Not Executed	Pending
TC40	TS09	Submit a valid feedback rating for advisory received	Rating: 4/5; Comment: 'Helpful advice'	Feedback is saved and confirmat ion	Not Executed	Pending

				message is shown		
TC41	TS09	Submit feedback with rating out of valid range (BVA - invalid)	Rating: 6 (valid range 1-5)	System rejects with 'Rating must be between 1 and 5'	Not Executed	Pending
TC42	TS10	Admin adds a new disease record to the database	Disease: 'Leaf Curl Virus' with treatment details	New record is saved and available for advisory matching	Not Executed	Pending
TC43	TS10	Admin edits an existing disease treatment recommendation	Update dosage for 'Powdery Mildew'	Updated details are reflected immediately in advisory results	Not Executed	Pending
TC44	TS10	Admin views analytics dashboard for regional disease trends	Filter: Last 30 days, Region: Tamil Nadu	Dashboard displays accurate charts of detected disease frequency	Not Executed	Pending
TC45	TS10	Non-admin user attempts to access admin	Login as farmer role, navigate to /admin	Access is denied with	Not Executed	Pending

		dashboard (Security)		'Unauthorized access' message		
TC46	TS11	Upload image while device is offline	Network: Disabled; Image ready to upload	Request is queued locally with 'Pending sync' status	Not Executed	Pending
TC47	TS11	Verify automatic sync when connectivity is restored	Network restored after being offline	Queued requests are automatically uploaded and processed	Not Executed	Pending
TC48	TS11	Verify app usability on 2G/low-bandwidth network (NFR7)	Network: 2G simulated	Core screens load with degraded but functional performance	Not Executed	Pending
TC49	TS12	Verify SMS notification is sent for completed diagnosis	Diagnosis completed for Farmer ID 1023	SMS with summary result is delivered within 1 minute	Not Executed	Pending

TC50	TS12	Verify push notification when app is in background	App state: Background	Push notification appears in device notification tray	Not Executed	Pending
TC51	TS13	Verify uploaded images are stored in encrypted form	Inspect storage of uploaded image files	Images are stored with encryption at rest	Not Executed	Pending
TC52	TS13	Verify session expires after prolonged inactivity	No user action for 30 minutes	Session times out and re-authentication is required	Not Executed	Pending
TC53	TS13	Attempt SQL injection in the disease search field (Security)	Search input: ' OR '1'='1	Input is sanitized; no unauthorized data is returned	Not Executed	Pending
TC54	TS14	Verify system response under 1,000 concurrent image uploads	Load test: 1,000 concurrent users	System processes requests without crash; response within acceptable limits	Not Executed	Pending

TC55	TS14	Verify system behavior at maximum supported concurrent load boundary (BVA)	Load test: exactly 10,000 concurrent requests	System remains stable and meets NFR5 scalability target	Not Executed	Pending
TC56	TS15	Validate detection accuracy against a labeled test dataset	100 pre-labeled disease images	Model achieves ≥ 85% precision and recall as per NFR6	Not Executed	Pending
TC57	TS15	Validate false-positive rate on healthy leaf samples	50 labeled healthy-leaf images	False positive rate remains within acceptable threshold	Not Executed	Pending

10. Traceability Matrix

The traceability matrix below maps each requirement-linked test scenario to its corresponding detailed test cases, demonstrating that every identified requirement has adequate test coverage.

Scenario ID	**Linked Requirement(s)**	**Scenario Description**	**Mapped Test Case IDs**
TS01	FR1	Verify farmer registration and login functionality	TC01, TC02, TC03, TC04, TC05, TC06, TC07, TC08

TS02	FR2	Verify crop image capture and upload functionality	TC09, TC10, TC11, TC12, TC13, TC14, TC15, TC16
TS03	FR3	Verify AI-based disease detection and confidence scoring	TC17, TC18, TC19, TC20, TC21, TC22, TC23
TS04	FR4	Verify advisory recommendation generation based on detected disease	TC24, TC25, TC26, TC27, TC28
TS05	FR5	Verify disease database search and view functionality	TC29, TC30, TC31
TS06	FR6	Verify weather-based alert notifications	TC32, TC33, TC34
TS07	FR7	Verify multilingual interface switching	TC35, TC36
TS08	FR8	Verify diagnosis history logging and report generation	TC37, TC38, TC39
TS09	FR9	Verify feedback and rating submission	TC40, TC41
TS10	FR10	Verify admin dashboard operations (user/database management, analytics)	TC42, TC43, TC44, TC45
TS11	FR11 / NFR10	Verify system behavior under offline / low-bandwidth conditions	TC46, TC47, TC48
TS12	FR12	Verify SMS and push notification delivery	TC49, TC50
TS13	NFR4	Verify data security and access control	TC51, TC52, TC53
TS14	NFR1 / NFR5	Verify system performance and scalability under load	TC54, TC55
TS15	NFR6	Verify AI model detection accuracy against known samples	TC56, TC57

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11. Entry and Exit Criteria

Defining clear entry and exit criteria ensures that testing begins only when the system and environment are ready, and concludes only when adequate quality and coverage have been demonstrated.

Entry Criteria	Exit Criteria
Requirements (FR/NFR) are reviewed and approved	All planned test cases have been executed at least once
Test environment (devices, network simulators, test accounts) is set up and available	No open Critical or High-severity defects remain unresolved
A stable build of the application/API is deployed to the test environment	At least 95% of designed test cases pass, with remaining failures documented and accepted
Test data (sample images, farmer accounts, disease records) is prepared	Traceability matrix confirms 100% requirement coverage has been executed
Test case design document (this document) is reviewed and signed off	Test summary report is prepared and shared with stakeholders

12. Risk Analysis and Mitigation

The following risks were identified as most relevant to the quality and field reliability of the platform, along with their likelihood, potential impact, and planned mitigation, which is reflected in the corresponding test cases.

Risk	Likelihood	Impact	Mitigation
AI model misclassifies disease due to poor image quality	High	High	Enforce image-quality checks before analysis; prompt farmer to retake unclear photos
Third-party weather API downtime	Medium	Medium	Cache last known forecast; degrade gracefully with a 'service unavailable' notice

Poor network connectivity in rural areas affects upload success	High	High	Implement offline queuing and automatic retry/sync (FR11, NFR10)
Unauthorized access to farmer data or admin functions	Low	High	Enforce role-based access control and encryption (NFR4); test with TC45, TC51-TC53
Low literacy users unable to navigate the app	Medium	Medium	Icon-driven, multilingual UI with usability testing (NFR2, NFR8)

13. Test Coverage Summary

Metric	Value
Total Functional Requirements	12
Total Non-Functional Requirements	10
Total Test Scenarios Identified	15
Total Detailed Test Cases Designed	57
Requirement Coverage	100% (all FR/NFR mapped to at least one scenario)
Test Design Techniques Applied	Equivalence Partitioning, BVA, Decision Table, State Transition

The designed test suite provides complete traceability from requirements through test scenarios to individual test cases. Normal, boundary, invalid, and exceptional conditions are covered for all major functional flows, and four recognized test design techniques (Equivalence Partitioning, Boundary Value Analysis, Decision Table Testing, and State Transition Testing) were systematically applied to strengthen defect-detection capability.

14. Conclusion

This Test Case Design document comprehensively addresses the requirements of the AI-Based Crop Disease Detection and Farmer Advisory Platform. By identifying functional and non-functional requirements, deriving test scenarios, applying multiple test design techniques, and mapping full traceability from requirements to test cases, the document establishes a strong foundation for the subsequent test execution phase. Executing this test suite will help verify the platform's correctness, reliability, security, and usability before deployment to farmers in the field.

15. Test Deliverables

- Test Case Design Document (this document), including requirements, scenarios, techniques, and detailed test cases.
- Traceability Matrix mapping requirements to test scenarios and test cases.
- Test Execution Log capturing Actual Result and Pass/Fail Status for each test case (to be completed during execution).
- Defect Report summarizing any issues identified during execution, with severity and status.
- Test Summary Report consolidating overall test coverage, pass/fail statistics, and residual risk at the end of the testing cycle.

16. Glossary of Terms

Term	**Definition**
BVA	Boundary Value Analysis - a test design technique that targets values at the edges of valid input ranges.
EP	Equivalence Partitioning - a technique that divides input data into partitions expected to be handled similarly.
FR / NFR	Functional Requirement / Non-Functional Requirement.
TC	Test Case - a documented set of steps, data, and expected results used to verify a specific behavior.
TS	Test Scenario - a high-level condition or feature area to be tested, broken down into one or more test cases.
CNN	Convolutional Neural Network - the deep-learning architecture used for image-based disease classification.
Confidence Score	A probability value output by the AI model indicating certainty of a disease prediction.